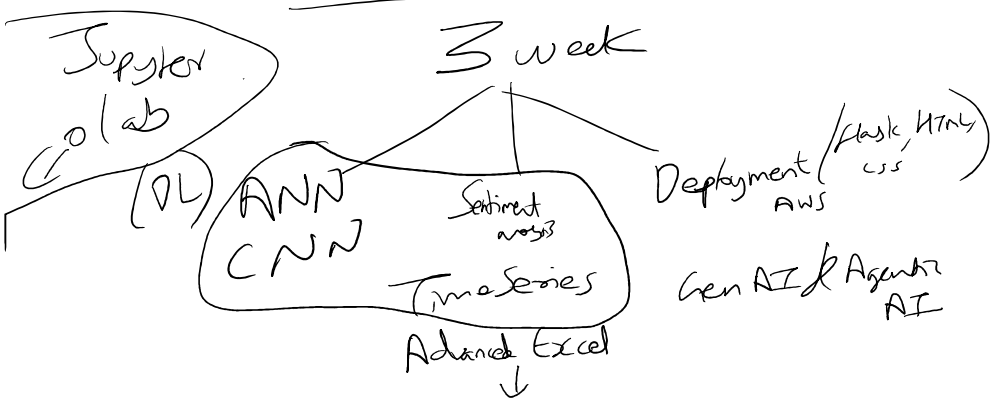
Monday, May 25, 2026 8:30 PM

ADS – Advance Data Science



ADS Day -1

Deep Learning

ANN - Artificial Neural Network

→ Deep learning is a subset/subfield of ML.

→ But it can work more advanced/complex than ML.

→ It can handle more data than ML



Main motive of AI → replicate human thinking

↓

Main motive of DL → replicate human thinking

↓

(Neural network) BRAIN (Biological Neurons)

↓

Artificial Neural Network
(Artificial Neurons)

→ ANN is the structure of Deep learning consisting of layers of neurons

→ Each layer carries a function

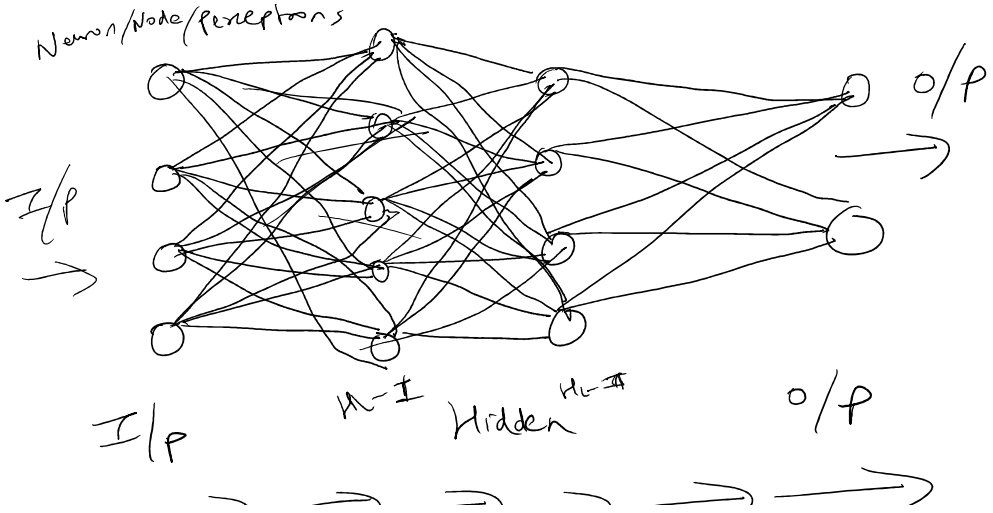
Architecture of ANN

A deeplearning n/w consist of 3 important types of layers:

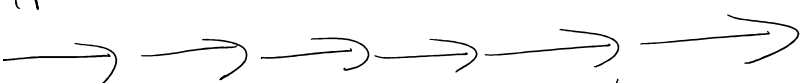
→ INPUT LAYER → Give i/p

→ HIDDEN LAYER → Processing, computation, learning
(outside world have no connection)

→ OUTPUT LAYER → Get o/p



I/p ... Hidden ...



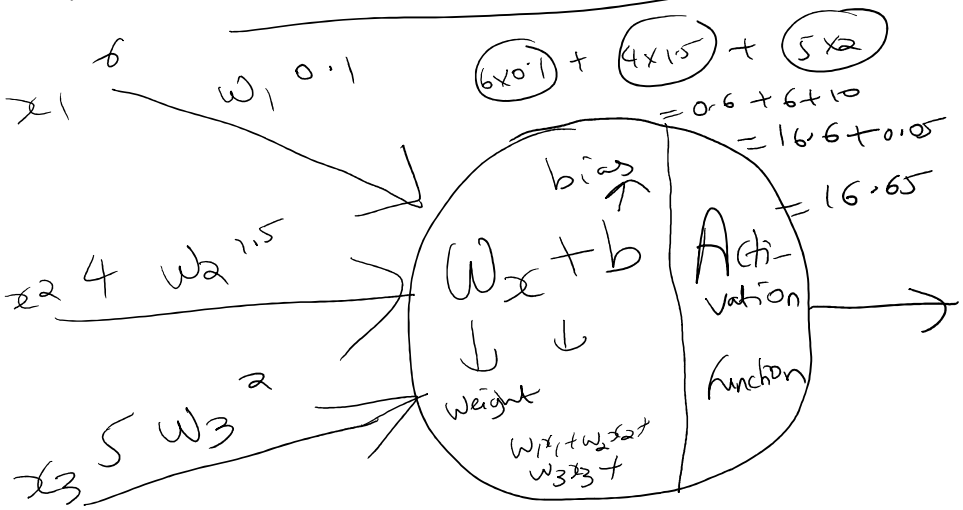
Data from i/p → hidden → o/p

↳ FORWARD PROPAGATION IN DEEP LEARNING

No: of neurons in I/p layer → No: of i/p features in our dataset

No: of neuron in o/p layer → No: of categories for classification

What's inside a neuron ?



WEIGHT

Weight → Strength of connection b/w neurons

Bias → Small constant value added by each neuron

Weight → Strength of connection b/w neurons

Bias → Small constant value added by each neuron

Activation function decides whether a neuron should be activated or not.

↓

Any non-linear function can be called as activation function

↓

Introduces non-linearity to our model

3 things inside a neuron

- WEIGHT
- BIAS
- ACTIVATION FUNCTION

Types of Activation functions

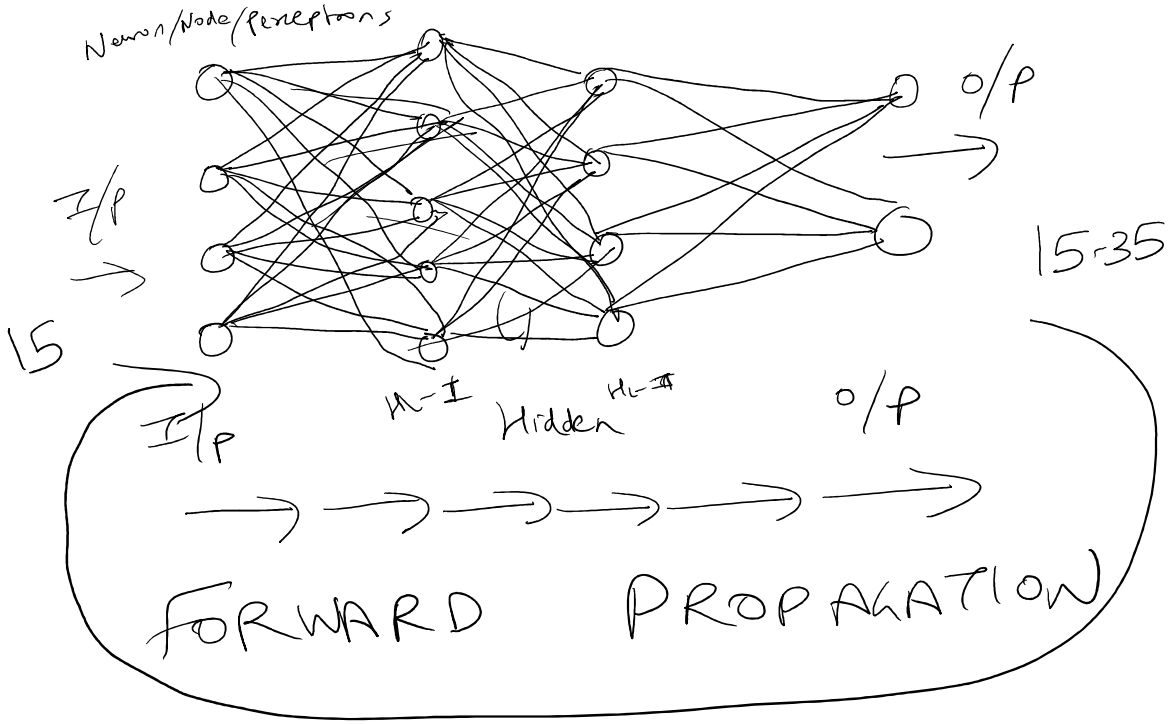
1 → SIGMOID → Binary classification

2 → SOFTMAX → Multi-class classification

3 → RELU → Rectified linear unit (Most used activation function) 0 to ∞

4 → tanh → -1 to 1 (hyperbolic tangent) (vanishing gradient problems)

(hyperbolic tangent) (vanishing gradient problems)



That repetitive reversal of movement from o/p to i/p

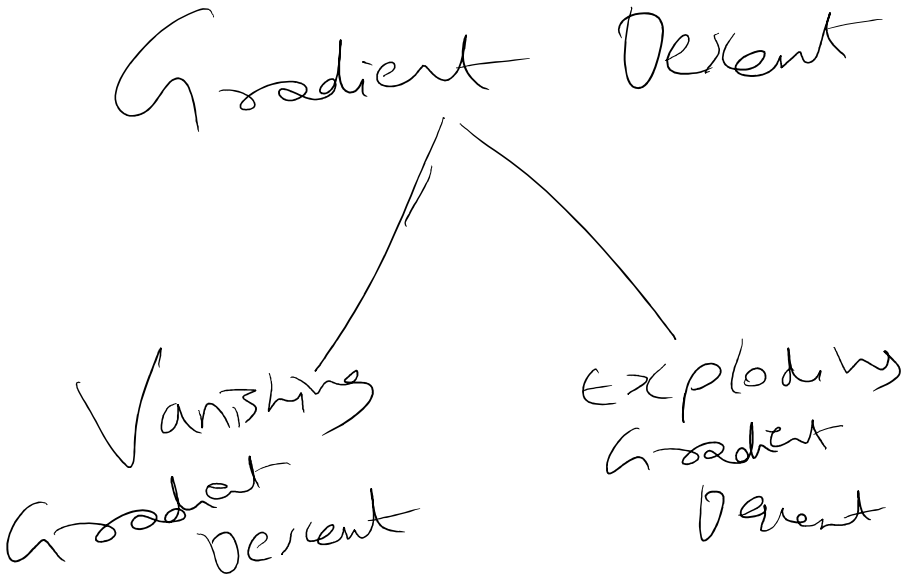
↓

BACKPROPAGATION

In BP, each layer of neural are updated with their weights & bias values such that errors & reduce

Gradient Descent

↳ technique used to find errors of each layer



20 ← 16 ← 15 ← 12 ← 10 ←

As the backpropagation goes from o/p to i/p, ERROR GOES ON INCREASING

EXPLODING GRADIENT

10 ← 12 ← 15 ← 16 ← 20 ←

ERROR GOES ON DIMINISHING

VANISHING GRADIENT